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1. Create a view named STAFF_SAL_VIEW to display staff id, name, monthly salary, and dept id from the STAFF table.

Code:

SQL> CREATE VIEW staff_sal_view_09399 AS

2 SELECT staff_id_09399, full_name_09399, monthly_salary_09399, dept_id_09399

3 FROM staff_09399;

```
SQL> CREATE VIEW staff_sal_view_09399 AS
  2  SELECT staff_id_09399, full_name_09399, monthly_salary_09399, dept_id_09399
  3  FROM staff_09399;
View created.
```

2. Create a view named MUMBAI_STAFF_VIEW to show details of employees working in departments located.

Output:

```
SQL> CREATE VIEW mumbai_staff_view_09399 AS  
2 SELECT *  
3 FROM staff_09399  
4 WHERE dept_id_09399 IN (SELECT dept_id_09399  
5 FROM department_09399  
6 WHERE location_09399 = 'Mumbai');
```

```
SQL> CREATE VIEW mumbai_staff_view_09399 AS  
2 SELECT *  
3 FROM staff_09399  
4 WHERE dept_id_09399 IN (SELECT dept_id_09399  
5 FROM department_09399  
6 WHERE location_09399 = 'Mumbai');  
  
View created.
```

3. Create a view named PAYBAND_SUMMARY that shows grade_level and the average of (min_pay + max_pay)/2 as average_band_value.

Output:

```
SQL> CREATE VIEW payband_summary_09399 AS  
2 SELECT grade_level_09399, (min_pay_09399 + max_pay_09399)/2 AS average_band_value_09399  
3 FROM payband_09399;
```

```
SQL> CREATE VIEW payband_summary_09399 AS  
2 SELECT grade_level_09399, (min_pay_09399 + max_pay_09399)/2 AS average_band_value_09399  
3 FROM payband_09399;  
  
View created.
```

4. Create an index named IDX_DEPT_ID on the column dept_id in the STAFF table.

Output:

```
SQL> CREATE INDEX idx_dept_id_09399
```

```
2 ON staff_09399 (dept_id_09399);
```

```
SQL> CREATE INDEX idx_dept_id_09399
2 ON staff_09399 (dept_id_09399);
Index created.
```

5. Create a composite index named IDX_JOB_SAL on the columns job_title and monthly_salary in the STAFF table.

Output:

```
SQL> CREATE INDEX idx_job_sal_09399
```

```
2 ON staff_09399 (job_title_09399, monthly_salary_09399);
```

```
SQL> CREATE INDEX idx_job_sal_09399
2 ON staff_09399 (job_title_09399, monthly_salary_09399);
Index created.
```

6. Add a new column named email_id VARCHAR2(40) to the STAFF table.

Output:

```
SQL> ALTER TABLE staff_09399
```

```
2 ADD (email_id_09399 VARCHAR2(40));
```

```
SQL> ALTER TABLE staff_09399
2 ADD (email_id_09399 VARCHAR2(40));
Table altered.
```

7. Modify the column monthly_salary in the STAFF table to increase its size to NUMBER(10,2).

Output:

```
SQL> ALTER TABLE staff_09399
```

```
2 MODIFY (monthly_salary_09399 NUMBER(10, 2));
```

```
SQL> ALTER TABLE staff_09399
2 MODIFY (monthly_salary_09399 NUMBER(10, 2));
Table altered.
```

8. Add a constraint to ensure commission_amt cannot be negative in the STAFF table.

Output:

```
SQL> ALTER TABLE staff_09399
```

```
2 ADD CONSTRAINT chk_comm_positive
```

```
3 CHECK (commission_amt_09399 >= 0);
```

```
SQL> ALTER TABLE staff_09399
2 ADD CONSTRAINT chk_comm_positive
3 CHECK (commission_amt_09399 >= 0);
Table altered.
```

9. Rename the column dept_name to department_name in the DEPARTMENT table.

Output:

```
SQL> ALTER TABLE department_09399
```

```
2 RENAME COLUMN dept_name_09399 TO department_name_09399;
```

```
SQL> ALTER TABLE department_09399
2 RENAME COLUMN dept_name_09399 TO department_name_09399;
Table altered.
```

10. Increase the monthly_salary of all employees in department 20 by 10%.

Output:

SQL> UPDATE staff_09399

2 SET monthly_salary_09399 = monthly_salary_09399 * 1.10

3 WHERE dept_id_09399 = 17;

```
SQL> UPDATE staff_09399
  2 SET monthly_salary_09399 = monthly_salary_09399 * 1.10
  3 WHERE dept_id_09399 = 17;
2 rows updated.
```

11. Set the commission_amt to 500 for all staff having job_title = 'CLERK'.

Output:

SQL> UPDATE staff_09399

2 SET commission_amt_09399 = 500

3 WHERE job_title_09399 = 'CLERK';

```
SQL> UPDATE staff_09399
  2 SET commission_amt_09399 = 500
  3 WHERE job_title_09399 = 'CLERK';
3 rows updated.
```

12. Change the location of department 30 to 'DELHI'.

Output:

SQL> UPDATE department_09399

2 SET location_09399 = 'DELHI'

3 WHERE dept_id_09399 = 17;

```
SQL> UPDATE department_09399
  2 SET location_09399 = 'DELHI'
  3 WHERE dept_id_09399 = 17;
1 row updated.
```

13. Display staff_id, full_name, and NVL(commission_amt, 0) as COMMISSION_VALUE for all employees.

Output:

```
SQL> set linesize 200
```

```
SQL> SELECT staff_id_09399, full_name_09399, NVL(commission_amt_09399, 0) AS  
commission_value
```

```
2 FROM staff_09399;
```

```
SQL> set pagesize 100
SQL> set linesize 200
SQL> SELECT staff_id_09399, full_name_09399, NVL(commission_amt_09399, 0) AS commission_value
2 FROM staff_09399;
```

STAFF_ID_09399	FULL_NAME_09399	COMMISSION_VALUE
11	ELEVEN	1111
1	ALEX	0
3	ALLEN	0
4	FOX	500
5	LION	1200
6	NASHRA	5000
7	MADINAH	500
8	KHADIJAH	1000
9	TAIBA	0
10	EAGLE	500
12	RIYA	0
13	MAQ	11000
14	GAUSIYA	6000
15	JOHN	500
17	ELON	0

15 rows selected.

14. Show full_name, monthly_salary, and total earning as (monthly_salary + commission_amt) for all employees.

Output:

```
SQL> SELECT full_name_09399, monthly_salary_09399, (monthly_salary_09399 +  
2 NVL(commission_amt_09399, 0))  
FROM staff_09399;
```

```
SQL> SELECT full_name_09399, monthly_salary_09399, (monthly_salary_09399 + NVL(commission_amt_09399, 0))  
2 FROM staff_09399;
```

FULL_NAME_09399	MONTHLY_SALARY_09399	(MONTHLY_SALARY_09399+NVL(COMMISSION_AMT_09399,0))
ELEVEN	12222	11111
ALEX	1100	1100
ALLEN	2000	2000
FOX	3500	3000
LION	5700	4500
NASHRA	11000	6000
MADINAH	11500	11000
KHADIJAH	11000	10000
TAIBA	8000	8000
EAGLE	70500	70000
RIYA	3000	3000
MAQ	23100	12100
GAUSIYA	18000	12000
JOHN	4100	3600
ELON	1100	1100

15 rows selected.

15. Delete all staff working in department 40.

Output:

```
SQL> DELETE FROM staff_09399  
2 WHERE dept_id_09399 = 40;
```

```
SQL> DELETE FROM staff_09399  
2 WHERE dept_id_09399 = 40;  
0 rows deleted.
```

16. Create a new table SAMPLE with 1 column “Serial no”. Once created add 2 rows in this table. Now Drop all data from this table but keep its structure

Output:

```
SQL> CREATE TABLE sample_09399 (serial_no NUMBER(5))
```

```
2 ;
```

Table created.

```
SQL> INSERT INTO sample_09399 (serial_no) VALUES (1);
```

1 row created.

```
SQL> INSERT INTO sample_09399 (serial_no) VALUES (2);
```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL>
```

```
SQL> CREATE TABLE sample_09399 (serial_no NUMBER(5))
2 ;

Table created.

SQL> INSERT INTO sample_09399 (serial_no) VALUES (1);

1 row created.

SQL> INSERT INTO sample_09399 (serial_no) VALUES (2);

1 row created.

SQL> COMMIT;

Commit complete.

SQL>
```